MTH 634

Quiz No.2

Solution by Sufyan JUTT

JUTT

MTH634 - Topology (Quiz No. 2) **Quiz Start Time: 08:37 PM, 03 December 2023**

Question # 1 of 10 (Start time: 08:37:04 PM, 03 December 2023) **Total Marks:** 1

Interior of a set A(denoted by Ao) which is a subset of a topological space X is the .....of all ..... subsets of X contained in A.

Select the correct option

- (●) intersection, open
- ( ) intersection, closed
- ( ) union,closed
- ( ) union, open

Click to Save Answer & Move to Next Question

SUTT

MTH634 - Topology (Quiz No. 2) Quiz Start Time: 08:37 PM, 03 December 2023

Question # 2 of 10 (Start time: 08:37:49 PM, 03 December 2023) **Total Marks:** 1

The following collection is a basis on R^2.

Select the correct option

Reload Math Equations

- (●) B = $\{(a, b) \times (c, d) \subseteq R^2 : a < b, c < d\}$
- ( ) $B = \{[a, b) \subseteq R : a < b\}$

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MTH634 - Topology (Quiz No. 2) Quiz Start Time: 08:37 PM, 03 December 2023

Question # 3 of 10 (Start time: 08:38:57 PM, 03 December 2023) **Total Marks:** 1

If A is a subset of a topological space X, then

Select the correct option

Reload Math Equations

- (●) exterior of A is an interior of A^c.
- ( ) exterior of A is an interior of A.

Click to Save Answer & Move to Next Question

MTH634 - Topology (Quiz No. 2) **Quiz Start Time: 08:37 PM, 03 December 2023**

Question # 4 of 10 (Start time: 08:39:46 PM, 03 December 2023) **Total Marks:** 1

Basis for indiscrete topology (ϕ, X) is.....

Select the correct option

Reload Math Equations

- (●) {X}
- ( ) empty set
- ( ) {0}
- ( ) $\{\phi, X\}$

Click to Save Answer & Move to Next Question

MTH634 - Topology (Quiz No. 2) **Quiz Start Time: 08:37 PM, 03 December 2023**

Question # 5 of 10 (Start time: 08:40:26 PM, 03 December 2023) **Total Marks:** 1

Consider the following topology on $X = \{a, b, c, d, e\}$:

$T = \{\emptyset, X, \{a\}, \{a, b\}, \{a, c, d\}, \{a, b, c, d\}, \{a, b, e\}\}$

The member of the relative topology on $A = \{a, b\}$ is

Select the correct option

Reload Math Equations

- ( ) {e}
- ( ) {c}
- (●) {a, b}
- ( ) none

Click to Save Answer & Move to Next Question

MTH634 - Topology (Quiz No. 2) Quiz Start Time: 08:37 PM, 03 December 2023

Question # 6 of 10 (Start time: 08:41:06 PM, 03 December 2023) **Total Marks:** 1

Consider the following topology on $X = \{a, b, c, d, e\}$:

$$T = \{\emptyset, X, \{a\}, \{a, b\}, \{a, c, d\}, \{a, b, c, d\}, \{a, b, e\}\}$$

The member of the relative topology on $A = \{a, b\}$ is

Select the correct option

Reload Math Equations

- ○ {c}
- ○ {a}
- ◉ none
- ○ {b}

Click to Save Answer & Move to Next Question

MTH634 - Topology (Quiz No. 2) Quiz Start Time: 08:37 PM, 03 December 2023

Question # 7 of 10 (Start time: 08:42:40 PM, 03 December 2023) **Total Marks:** 1

The set of all singletons forms a basis for ..... topology.

Select the correct option

- ◉ discrete
- ○ lower limit
- ○ indiscrete
- ○ co finite

Click to Save Answer & Move to Next Question

MTH634 - Topology (Quiz No. 2) **Quiz Start Time: 08:37 PM, 03 December 2023**

Question # 8 of 10 (Start time: 08:43:24 PM, 03 December 2023) **Total Marks:** 1

If A is a subset of a topological space X, then

Select the correct option

Reload Math Equations

- [ ] $Ext(A)$ is the smallest open subset of A^c.
- [x] $Ext(A)$ is the largest open subset of A^c.

Click to Save Answer & Move to Next Question

MTH634 - Topology (Quiz No. 2) **Quiz Start Time: 08:37 PM, 03 December 2023**

Question # 9 of 10 (Start time: 08:44:09 PM, 03 December 2023) **Total Marks:** 1

If A is a subset of a topological space X, then

Select the correct option

Reload Math Equations

- [ ] $Ext(A) \cup Int(A) = \phi$
- [x] $Ext(A) \cap Int(A) = \phi$.

Click to Save Answer & Move to Next Question

MTH634 - Topology (Quiz No. 2) Quiz Start Time: 08:37 PM, 03 December 2023

Question # 10 of 10 (Start time: 08:45:03 PM, 03 December 2023) Total Marks: 1

Let A be a non empty proper subset of an indiscrete topological space X. The boundary of A is

Select the correct option

Reload Math Equations

- (●) X
- ( ) $\emptyset$

Click to Save Answer & Move to Next Question

Sufyan